

(Ad hoc)

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$$R_i = \underbrace{\alpha_i}_{\text{(const)}} + \underbrace{\beta_i}_{\text{(slope)}} R_m + \underbrace{\epsilon_i}_{\text{(idiosyncratic component)}}$$

(asset return) (sensitivity) (market return)

systematic risk factor.

$$\text{Corr}(\epsilon_i, R_m) = 0$$

$$\text{Corr}(\epsilon_i, \epsilon_j) = 0$$

$$\epsilon_i \sim N(0, \sigma_i^2)$$

$$\mathbb{E}[R_i] = \alpha_i + \beta_i \mathbb{E}[R_m]$$

$$\underbrace{V[R_i]}_{\text{(total risk)}} = \underbrace{\beta_i^2 V[R_m]}_{\text{(systematic risk)}} + \underbrace{V[\epsilon_i]}_{\text{(idiosyncratic risk)}} = \beta_i^2 \sigma_m^2 + \sigma_i^2$$

Portfolio of N assets.

$$\begin{aligned} R_p &= \sum w_i R_i = \sum w_i (\alpha_i + \beta_i R_m + \epsilon_i) \\ &= (\sum w_i \alpha_i) + (\sum w_i \beta_i) R_m + (\sum w_i \epsilon_i) \\ &= \bar{\alpha} + \bar{\beta} R_m + \sum w_i \epsilon_i \quad \text{Cov}(\epsilon_i, \epsilon_j) = 0 \end{aligned}$$

$$E[R_p] = \bar{\alpha} + \bar{\beta} \mu_m \quad (\mu_i, \sigma_i)$$

$$V[R_p] = \bar{\beta}^2 \sigma_m^2 + \sum w_i^2 \sigma_{\epsilon_i}^2$$

$$\begin{aligned} \text{Cov}(R_i, R_j) &= \text{Cov}(\alpha_i + \beta_i R_m + \epsilon_i, \alpha_j + \beta_j R_m + \epsilon_j) \\ &= \beta_i \beta_j \sigma_m^2 \end{aligned}$$

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\sigma_i \cdot \sigma_j} = \frac{\beta_i \beta_j \sigma_m^2}{\sigma_i \cdot \sigma_j} = \frac{\rho_{i,m} \left(\frac{\sigma_i}{\sigma_m} \right) \cdot \rho_{j,m} \left(\frac{\sigma_j}{\sigma_m} \right) \sigma_m^2}{\sigma_i \sigma_j} = \rho_{i,m} \times \rho_{j,m}$$

$$\begin{aligned} \beta_i &= \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \frac{\rho_{i,m} \times \sigma_i \times \sigma_m}{\sigma_m^2} \\ &= \rho_{i,m} \times \left(\frac{\sigma_i}{\sigma_m} \right) \end{aligned}$$

$$\begin{aligned} y^f &= \alpha + \beta x^f + \epsilon \\ \hat{\beta} &= \frac{\text{Cov}(x, y)}{\sigma_x^2} \end{aligned}$$

$$r_i = f_i(r_n) + \sqrt{1-f_i^2} \times \epsilon_i$$

$N \rightarrow \infty$

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$$\bar{\alpha} = \sum w_i \alpha_i = \sum \frac{1}{N} \times \alpha_i = \alpha$$

$$\begin{aligned} E[R_p] &= \bar{\alpha} + \bar{\beta} \mu_m \\ V[R_p] &= \bar{\beta}^2 \sigma_m^2 + \sum w_i^2 \sigma_i^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} E[R_p] &= \bar{\alpha} + \bar{\beta} \mu_m \\ V[R_p] &= \bar{\beta}^2 \sigma_m^2 + \sum w_i^2 \sigma_i^2 \end{aligned}} \right\} \text{portfolio}$$

homogeneous portfolio, $w_i = \frac{1}{N}$, $\sigma_i = \sigma$, $\beta_i = \beta$, $\alpha_i = \alpha$

$$E[R_p] = \alpha + \beta \mu_m, \quad E[R_i] = \alpha + \beta \mu_m$$

$$V[R_p] = \beta^2 \sigma_m^2 + \sum \frac{1}{N} \times \sigma^2 = \beta^2 \sigma_m^2 + \frac{\sigma^2}{N}$$

$N \rightarrow \infty$, $V[R_p] = \beta^2 \sigma_m^2$ ✓
 $\sigma[R_p] = \beta \sigma_m$ ✓

